Algebra 1 Name____________________
Unit 9 Date_____________________
Lesson 8 Homework Period___________________

Kalen's work when solving the equation $y = 7(x-5)^2 + 9$ for x is shown. Use Kalen's work to answer questions 1 and 2.

$$y = 7(x-5)^2 + 9$$
$$y - 9 = 7(x-5)^2 + 9 - 9$$
$$(y-9) \div 7 = 7(x-5)^2 \div 7$$
$$\sqrt{7(y-9)} = \sqrt{(x-5)^2}$$
$$\sqrt{7(y-9)} = x - 5$$
$$\sqrt{7(y-9)} + 5 = x$$

1. Circle the step where Kalen made an error.

2. Correct Kalen's error and the steps that follow to show how to correctly isolate x in the equation. Use the space provided above to show the correct steps.

Solve each equation for the designated variable. Show your work.

3. Solve $y = 4x^2 + px + 40$ for p.	4. Solve $y = \frac{1}{2}(x-4)^2 + r$ for r.

5. Solve $4(x-2)(x+w)=y$ for w.	6. Solve $y=x^2-12x+3m$ for m.
7. Solve $y=q(x+2)^2-1$ for q.	8. Solve $y=a(x-2)(3x-4)$ for a.
9. Solve $y=-3(x-2)^2+4$ for x.	10. Solve $y=ax^2+2x-9$ for a.